

DNS 8803 Project Proposal

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[Introduction]

As data centers scale to support AI and microservice workloads, a debate has emerged over whether Transmission Control Protocol (TCP) remains well suited for modern data center traffic. Modern microservices and RPC-style workloads rely on short, latency-sensitive messaging as a defining characteristic. Proponents of message-oriented protocols like Homa argue that TCP's connection-oriented design causes head-of-line blocking, which limits scalability and increases tail latency. This project aims to evaluate whether a carefully tuned TCP stack can meet or exceed Homa's performance in CloudLab testbeds under realistic traffic patterns and loads; thereby challenging the assumption that TCP must be replaced for next-generation data center communication.

[Summary of most relevant papers]

[Ousterhout, *It's Time to Replace TCP in the Datacenter* \(2022\).](#)

This paper argues that TCP's stream-oriented, connection-based design is poorly suited for modern data center workloads dominated by short, latency-sensitive RPCs. It claims that TCP induces transport-layer head-of-line blocking when independent messages are multiplexed over a single connection, leading to elevated tail latency under realistic load. The paper proposes Homa, a receiver-driven, message-oriented transport protocol, and presents experimental results suggesting that Homa significantly outperforms TCP in terms of latency and throughput for RPC-style workloads. However, Pepelnjak's critique in [Is It Time to Replace TCP in Data Centers?](#) argues that these results overestimate the practical impact of TCP's limitations by relying on experimental assumptions that are uncommon in production systems, such as multiplexing unrelated messages over a single TCP connection, and by overlooking well-known mitigations and existing alternatives.

[Scharf and Kiesel, *Head-of-Line Blocking in TCP and SCTP: Analysis and Measurements*](#)

This paper provides a systematic analysis of head-of-line blocking arising from TCP's single ordered byte stream when multiple independent messages share a connection. Through analytical modeling and experiments, it demonstrates how transport-layer ordering constraints can delay unrelated data and shows that multi-stream transport designs, such as SCTP, can alleviate these effects. The study helps distinguish structural transport-layer head-of-line blocking from congestion-related latency, informing experimental comparisons between stream-based and message-oriented transports.

[Proposed Idea]

This project investigates whether a carefully tuned TCP stack can match the performance, or even outperform, Homa when evaluated under realistic workloads. We will build our experimental framework around the open-source HomaModule GitHub repository, which provides a Linux kernel implementation of the Homa transport protocol along with benchmarking tools for message-oriented RPC workloads. Using CloudLab, we will deploy controlled multi-node testbeds with configurable topologies, link speeds, and background traffic to fairly compare Homa against a properly setup (vanilla) TCP, as well as RDMA

– serving as a baseline test metric. By generating realistic mixes of short messages and competing flows, this project aims to empirically assess whether HOMA’s claims against TCP hold any merit.

GITHUB REPO: <https://github.gatech.edu/sahmed393/8803-Project-TCP-VS-HOMA->

CLOUDLAB RESOURCES:

NODES:

8-10 Worder Nodes (These will generate and receive traffic)

2-3 Switches (Used to create a simulated datacenter topology)

Node Types:

Ideal Node Type

- c6620 (100gb NIC)

Realistic node Type

- m510 (10gb NIC)

Projected Max ram and storage use (most likely experiment will be very comfortably under this)

- 16gb ram

- 20-30 gb storage

Operating System: Linux, Probably Ubuntu